Supplemental Results Discussion for MLPerf Inference v6.0

MLCommons Supplemental Discussion

MLPerf Inference v6.0 Results Discussion

The submitting organizations provided the following 300-word descriptions as a supplement to help the public understand their MLCommons® MLPerf® Inference v6.0 submissions and results. The statements **do not reflect the opinions or views of MLCommons.**

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AMD

For the MLPerf Inference v6.0 round, AMD is pleased to share new results showcasing expanded model coverage and continued software and system innovation on AMD Instinct™ GPUs. New in this submission, AMD and its OEMs submitted large scale multi-node inference results for some of the most popular AI models. The submission set includes configurations spanning AMD Instinct MI355X, MI350X, and MI325X, and MI300X GPUs (delivered via OEMs) to provide a representative view of performance across the AMD datacenter GPU portfolio under MLCommons rules. Submissions span the MLPerf Server, Offline, and Interactive scenarios, reflecting both request-driven serving and throughput-oriented batch inference.

AMD has achieved a major milestone in achieving over one million tokens per second on multi-node Llama2-70B in both Server and Offline and in GPT-OSS-120b Offline tests. This demonstrates strength of AMD software and hardware solutions enabling strong sustained throughput across multiple inference modes in large scale AI model serving.

This round also introduces two new MLPerf model submissions on AMD Instinct hardware: GPT-OSS-120B and WAN2.2 Text-to-Video. This is the first time that AMD has submitted highly performant implementations for newly introduced models demonstrating increased velocity performance improvements. An Open division submission featuring Llama 3.1-405B further broadens coverage for large dense models and emerging usage patterns.

Ecosystem participation remains strong, with nine OEMs submitting results on AMD Instinct platforms: Cisco (MI350X), Dell (MI300X, MI355X), MangoBoost (MI300X, MI325X, MI355X; including heterogeneous), GigaComputing (MI355X), HPE (MI355X), MiTAC (MI350X), Oracle (MI355X), Supermicro (MI350X, MI355X), and Red Hat (MI350X).

Together, these MLPerf v6.0 Inference submissions reinforce a continued commitment to transparent benchmarking, ecosystem collaboration, and progress across next-generation inference workloads.

ASUSTeK

ASUS Demonstrates Outstanding Performance in MLCommons MLPerf Inference Benchmark
Commitment to Open Collaboration Drives Industry-Leading AI Infrastructure Results

ASUS today announced its continued participation and exceptional results in the MLCommons MLPerf Inference benchmark, one of the most respected and rigorous AI performance evaluations in the industry. This milestone underscores ASUS's ongoing dedication to advancing AI computing standards on a global scale.

In this latest submission cycle, ASUS delivered a complete and comprehensive set of benchmark results across multiple inference categories, reflecting the company's commitment to full transparency and thorough performance validation. By consistently contributing complete benchmark suites rather than selective submissions, ASUS demonstrates both the versatility and reliability of its server and workstation platforms under real-world AI workloads.

Beyond performance numbers, ASUS places great emphasis on building a collaborative and mutually beneficial ecosystem within the MLCommons community. By actively engaging with industry partners, hardware vendors, and research institutions through the MLCommons framework, ASUS fosters an environment where shared progress elevates the entire AI infrastructure landscape — embodying a philosophy of co-creation and shared success.

Through the continuous publication of benchmark data across successive MLPerf rounds, ASUS systems are progressively aligning with and exceeding international AI performance standards. Each submission cycle serves as both a validation checkpoint and a development milestone, enabling ASUS engineering teams to refine hardware configurations, optimize software stacks, and ensure that ASUS platforms meet the evolving demands of enterprise AI deployments worldwide.

“Our participation in MLPerf is not just about scores — it is about accountability, collaboration, and pushing the boundaries of what AI infrastructure can achieve,” said a spokesperson from ASUS. “We remain committed to contributing meaningfully to the global AI community.”

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Cisco

Cisco AI Infrastructure: Scalable Performance for Inference

For MLPerf™ Inference v6.0, Cisco demonstrates the exceptional versatility and scalability of its AI infrastructure in the Datacenter Closed division. This submission spans a diverse range of compute engines—including the latest NVIDIA B300, RTX Pro 6000 and H200 GPUs, AMD Instinct™ MI350X accelerators, and Intel Xeon CPUs—proven across our C-Series and modular X-Series platforms.

Validated Multi-Node Scaling with Silicon One

A defining highlight of this submission is the validation of multi-node cluster performance. By comparing a single-node Cisco UCS with 8x AMD Instinct™ MI350X against a 16x MI350X multi-node configuration, Cisco demonstrates near-linear scaling. These nodes are interconnected by the Cisco Silicon One G200-based Ethernet fabric. Utilizing **Intelligent Packet Flow** technology, the G200 optimizes path utilization and ensures the lossless, high-bandwidth connectivity required to maintain peak throughput as organizations scale from individual nodes to distributed inference clusters.

Advanced AI Platforms and Modular Innovation

Cisco continues to simplify AI infrastructure with the launch of two advanced platforms featured in this submission:

Cisco UCS C880A M8 Rack Server: Based on the NVIDIA HGX architecture, this high-density, air-cooled platform is engineered for the most demanding AI and HPC workloads, showcased here with 8x NVIDIA B300 SXM GPUs.

Cisco UCS X580p PCIe Node: Extending the modular UCS X-Series, the X580p delivers high-performance GPU support for the NVIDIA H200 NVL and RTX Pro 6000, providing the flexibility needed for generative AI fine-tuning and inference.

Versatile Performance: The submission also features the C845A M8 with 8x H200 NVL GPUs and the C240 M8 powered by the Intel Xeon 6787P (Granite Rapids) processor, offering optimized performance for diverse inference tasks.

By combining the scale-out reliability of Cisco Silicon One with the modularity of UCS, Cisco delivers a validated, full-stack infrastructure for deploying AI at scale.

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Coreweave

In the latest MLPerf® Inference v6.0, CoreWeave participated in the Datacenter Closed division with NVIDIA GB200 NVL72 and NVIDIA GB300 NVL72, two of the newest GPU generations. CoreWeave's submissions reflect NVIDIA's reference configurations as a verified, production-ready baseline with some of the most demanding and popular reasoning models today: DeepSeek-R1 and GPT-OSS-120B.

Across CoreWeave submissions in this round, GB300 NVL72 delivered the strongest DeepSeek-R1 performance, leading the portfolio in offline throughput and per-GPU efficiency. In six months, server throughput effectively doubled versus MLPerf v5.1 on the same hardware footprint, reflecting rapid iteration and innovation. GB200 NVL72 followed closely, with strong DeepSeek-R1 results in both offline and server scenarios, demonstrating standout throughput with GPT-OSS-120B.

On both Blackwell Ultra GPU generations, CoreWeave delivered reliable inference on large-scale reasoning LLMs, including DeepSeek-R1's sparse Mixture-of-Experts architecture where efficient serving depends on dynamic expert routing across GPUs and high bandwidth inter-node communication.

CoreWeave’s performance benefits from continuous fleet-wide health monitoring paired with centralized benchmarking that ties system behavior directly to sustained tokens-per-second outcomes. Topology-aware Slurm on Kubernetes (SUNK) scheduling keeps inference jobs within high bandwidth NVLink domains on GB200 and GB300 systems, while CoreWeave Mission Control adds rack-to-cluster lifecycle management and deep operational visibility. An object storage plus node-local cache data path accelerates model startup, reduces load times, and stabilizes throughput at scale.

CoreWeave is built from the ground up to translate benchmark results like MLPerf into sustained, predictable throughput under real production traffic, which unfolds at the workload level. For the most demanding AI enterprises, labs, and AI-native startups with performance-sensitive workloads, CoreWeave delivers dedicated inference optimization and fine-tuning to the specific models, batch profiles, and latency requirements for each use case.

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Dell Technologies

Dell Technologies today announced a broad OEM submission portfolio in **MLPerf™ Inference v6.0**, highlighting its continued focus on validated, enterprise AI infrastructure. Across the Datacenter Closed and Edge categories, Dell delivered broad benchmark coverage spanning GPU-accelerated and CPU-only platforms—providing press, analysts, and customers with transparent, standards-based performance data across diverse AI deployment models.

Dell benchmarked multiple flagship platforms, including the **PowerEdge XE9680**, **PowerEdge XE9780/XE9785**, **PowerEdge XE7740/XE7745**, and the CPU-only **PowerEdge R770 (2-socket Granite Rapids)**. Submissions covered large language model inference, recommendation systems, computer vision, and additional datacenter workloads, along with YOLO object detection in the MLPerf Edge category—extending validated performance from core datacenter environments to distributed edge deployments.

With NVIDIA GPUs, Dell submitted results using B200/B300 GPUs in the XE9780 family and H200 NVL and RTX Pro 6000 GPUs in the XE7740/XE7745 platforms. These results demonstrate scalable inference performance across both liquid-cooled and air-cooled configurations, offering customers flexibility in performance density, power efficiency, and deployment architecture.

With AMD accelerators, Dell submitted results on the PowerEdge XE9680 (8× AMD Instinct MI300X) and PowerEdge XE9785 (8× AMD Instinct MI355X), delivering strong performance across transformer and recommendation benchmarks and validating support for next-generation accelerator architectures.

Dell also provided CPU-only baselines with the PowerEdge R770 powered by Intel Xeon 6787P and 6979P processors, enabling meaningful heterogeneous performance comparisons.

Together, these MLPerf Inference v6.0 results underscore Dell Technologies' commitment to rigorous, transparent benchmarking through MLCommons—strengthening MLPerf as a trusted, repeatable industry standard for evaluating AI inference performance.

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GATEOverflow

As part of our mission to make **MLPerf Inference** benchmarking more practical and reproducible on everyday hardware, we've successfully run submissions on **NVIDIA GPUs** and x86 **CPUs** — leveraging implementations from **NVIDIA and MLCommons**. We are happy to have submitted results for the newly created Yolo benchmark.

To simplify reproducibility at scale, we've also enhanced the **MLCFlow automation framework** from MLCommons, which powered all of our submissions and ensures **scalable, reliable performance benchmarking**.

We're excited to collaborate with partners who share our vision of making **MLPerf benchmarking accessible, repeatable, and impactful** across diverse hardware platforms.

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GigaComputing

Giga Computing is a GIGABYTE subsidiary that serves as the company's enterprise division responsible for designing, manufacturing, testing, and selling GIGABYTE server products. Giga Computing as one of the founding members (of MLCommons contributors) has continually contributed diverse systems in testing for each round of MLPerf Training and MLPerf Inference. We hope these results can allow academia, global technology providers, and others to have better expectations of what systems can achieve to drive timely insights and results.

For MLPerf Inference 6.0, three GIGABYTE systems were selected to reflect the broad choices of accelerated computing systems in the market today. The two 8U systems were optimized for airflow and performance using 8-GPU OAM and HGX baseboards. And one node of a eighteen compute node system (for the NVIDIA GB300 NVL72) was also introduced to testing to show performance of a Arm-based CPU.

- GIGABYTE G893-ZX1-AAX4, using:
  - 2x AMD EPYC 9575F (64 core) processors
  - AMD Instinct MI355X
- GIGABYTE G894-SD3-AAX7, using:
  - 2x Intel Xeon 6767P (64 core) processors
  - NVIDIA HGX B300
- GIGABYTE Compute Node for NVIDIA GB300 NVL72
  - 2x NVIDIA GB300 Grace Blackwell Ultra Superchip
  - 4x Blackwell Ultra GPUs
  - 2x Grace CPUs

To learn more about our solutions, visit: https://www.gigabyte.com/Enterprise and https://www.gigacomputing.com/en/

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HPE

MLPerf Inference v6.0 marks a milestone for HPE as the 10th round of results it has published in MLCommons MLPerf Inference, and marks more than 600 inference results submitted by HPE since v1.0. HPE thanks MLCommons and our partners for creating trusted and meaningful AI benchmark suites for the industry.

This round, HPE published 86 results featuring 14 configurations of HPE ProLiant Compute servers using NVIDIA Blackwell and AMD Instinct GPUs. This portfolio demonstrates why HPE ProLiant Compute is well-suited for a broad set of AI workloads, including large language models (LLMs), generative AI (Gen AI), speech-to-text, and vision applications for data center and edge inference. HPE also submitted a configuration using the new NVIDIA RTX PRO 4500 in this release.

<u>HPE servers for datacenter workloads:</u>
HPE ProLiant Compute XD685 with AMD EPYC CPUs delivered HPE's highest performance result for single server inference. HPE showcased two HPE ProLiant Compute XD685 direct liquid-cooling (DLC) configurations for highly-scalable, large data center inference – one with 8x NVIDIA B200 192GB GPUs, and another with AMD Instinct™ MI355X 288GB GPUs.

HPE ProLiant Compute DL380a Gen12 with Intel Xeon CPUs support up to 10 NVIDIA RTX PRO 6000 Blackwell Server Edition 96GB GPUs for medium-sized AI models on separate GPUs.

HPE ProLiant DL385 Gen11 with AMD EPYC CPUs support up to 4 NVIDIA RTX PRO 6000 GPUs when total cost of ownership (TCO) and performance per watt is a key decision factor.

<u>HPE servers for edge workloads:</u>
HPE ProLiant Compute DL380 Gen12 with Intel Xeon CPUs support up to 5 NVIDIA RTX PRO 4500 with 32GB memory each, which is useful when implementing small AI models for data center or edge applications.

HPE ProLiant DL145 Gen11 with AMD EPYC CPUs is a low-profile server suited for edge inference and supports one NVIDIA RTX PRO 4500.

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Intel

Intel is excited to share MLPerf 6.0 results across our portfolio of AI solutions led by Intel® Arc™ Pro B-Series GPUs – including the newly released Intel® Arc™ Pro B70 – and Intel® Xeon® 6 CPUs. Intel's AI Systems demonstrate accessible AI workload solutions across high-end workstations, datacenter, and edge.

Configuring a system with 4 Intel® Arc™ Pro B70 or B65 GPUs provides 128GB of VRAM, enabling 120b parameter models like GPT-OSS-120B alongside Llama2-70B, Llama3.1-8B, and Whisper to concurrent users. Intel® Arc™ Pro B70 GPU shows up to 1.8x performance gains over Intel® Arc™ Pro B60. Our results also demonstrate continuous improvements to our SW stack, achieving up to 1.18x improvement over MLPerf v5.1 using the same cards. An open, containerized companion software stack integrates drivers, frameworks, and tools to efficiently scale optimized AI inference workloads from single-node systems to multi-GPU enterprise deployments. The Intel® Arc™ Pro B70 GPU will be available from Intel and partners - either standalone or as part of OEM pre-built and validated inference workstations.

Intel remains the only vendor submitting server CPU results to MLPerf, reinforcing its CPU inferencing excellence. MLPerf v6.0 results demonstrate that Intel® Xeon® 6 delivers up to 1.9x improvement in AI performance over its previous generation, driven by higher core counts, expanded AMX acceleration, and greater memory bandwidth. Xeon continues to feature prominently as the host CPU for AI accelerated systems, with over 50% of MLPerf 6.0 submissions leveraging Xeon processors.

The combination of Intel® Xeon® 6 and Intel® Arc™ Pro B-Series GPUs represent our investment to expand customer choice and value, offering real-world solutions that address both LLMs as well as traditional machine learning workloads. Intel also supported partners - Cisco, Dell Technologies, Lenovo, Quanta, and Supermicro - with benchmark submissions on their Xeon® 6 based products.

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Inventec Corporation

Inventec, a global leader in high-performance computing (HPC) and AI server solutions, sets new AI performance milestones in MLPerf® Inference v6.0 Benchmarks. Featuring the flagship **P9000AG7** server, Inventec continues to redefine the boundaries of AI processing power, staying true to its corporate mission: “Inventing today, inspiring tomorrow.”

The Inventec **P9000AG7** is a high-density server purpose-built for generative AI and LLMs. It features dual **AMD EPYC™ 9005** processors and **NVIDIA Blackwell™ B200 GPUs**, achieving elite performance through a sustainable, advanced air-cooling design. Key innovations of Inventec **P9000AG7** include an optimized switchboard for seamless **NVLink™** and **NVIDIA GPUDirect® RDMA** connectivity, high-speed **DDR5 DIMM (6400 MT/s)**, up to 12 U.2 NVMe SSDs for rapid data access, and a unique **decoupled 12V and 54V power source design** that boosts efficiency. Furthermore, its modular build ensures easy scalability and serviceability for modern data centers.

The results in MLPerf® Inference v6.0 validate Inventec's commitment to engineering excellence. By combining NVIDIA’s cutting-edge Blackwell™ platform with thermal and power innovations, Inventec is providing the foundation for the next generation of AI.

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KRAI

Founded in 2020 in Cambridge, UK ("The Silicon Fen"), KRAI is a purveyor of premium benchmarking and optimization solutions for AI Systems. The KRAI team is proud to be a Founding Member of MLCommons, having participated in every MLPerf Inference round and more than half of MLPerf Training rounds to date.

Highlighted in this round are KRAI's results on the newly introduced Text-to-Video Generative AI benchmark using KISS-V, KRAI Inference Serving Solution for Video, on HPE Cray XD670 servers with 8x NVIDIA H200 GPUs.

In the Closed division, KRAI achieved an exceptional under-one-minute latency of 57.4 seconds per video.

In the Open division, KRAI achieved an even lower latency of 43.9 seconds per video using KISS-V's "Fast Mode", while meeting the quality constraints of the Closed division.

We cordially thank Hewlett Packard Enterprise for their long term partnership, which allows us to continue pushing the boundaries of performance benchmarking and optimization.

Lambda

NVIDIA Blackwell Architecture and MoE kernel optimizations enable lightning-fast inference

Lambda partnered with NVIDIA for this round of MLPerf® Inference submissions, running benchmarks on [Lambda’s Cloud] infrastructure equipped with two NVIDIA Blackwell systems: NVIDIA B200 GPUs each with 180GB SXM memory, 112 CPU cores, 2.9 TB RAM, and 28 TB SSD, as well as our newest accelerator for the first time during MLPerf inference, NVIDIA GB300s with 279GB HBM3e memory and 36 Grace CPUs.

Our inference benchmarks with the GPT-OSS-120B and Llama 3.1-8B models on the NVIDIA GB300 chips deliver extremely fast throughout, showcasing a sizable improvement in speed over the NVIDIA B200.

Additionally, this round of inference marks the first Open Division submission for Lambda. Using stack optimizations from our ML experts, we observed significant latency reductions in time-to-first-token (TTFT) while increasing throughput and maintaining accuracy for the

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GPT-OSS-120B model. This was achieved by balancing the routing of tokens in the mixture-of-experts (MoE) architecture.

The throughput achieved on NVIDIA GB300 systems validates that our infrastructure delivers excellent performance for demanding inference workloads. Furthermore, our Open Division submission proves that Lambda’s team of expert engineers is ready to help customers optimize their workloads to hit state-of-the-art efficiency. With Lambda, AI teams get proven speed, availability for on-demand instances, and the professional support required to deploy inference at scale.

About Lambda

Lambda, The Superintelligence Cloud, is a leader in AI cloud infrastructure serving tens of thousands of customers. Our customers range from AI researchers to enterprises and hyperscalers.

Lambda's mission is to make compute as ubiquitous as electricity and give everyone the power of superintelligence. One person, one GPU.

Founded in 2012 by machine learning engineers published at NeurIPS and ICCV, Lambda builds supercomputers for AI training and inference.

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Lenovo

Lenovo delivers smarter technology for all, for hardware, software and more importantly solutions that customers have come to know. Being smarter requires the research, testing and benchmarking that MLPerf Inference v6.0 provides. Allowing you to see first-hand how delivering smarter technology translates into tangible performance metrics.

Since partnering with MLCommons, Lenovo has been able to showcase such results quarterly in the MLPerf benchmarking. The insights gained from these benchmarks are one side of the benefit, whereas the other side allows us to constantly be improving our own technology for our customers based on our benchmarks by closely working with our partners in optimizing the overall performance.

We are excited to announce with our partners NVIDIA and Intel, that we competed on multiple important AI tasks such as Large Language Model, Summarization, Text Generation, Image Classification, Medical Image Segmentation, Speech-to-text, Node classification and Natural Language Processing running on our ThinkSystem SR680a V4, ThinkSystem SR675i V3, ThinkSystem SR650 V4, ThinkEdge SE100 and ThinkEdge SE455i V3. These partnerships have allowed us to consistently achieve improving results.

This partnership with MLCommons is key to the growth of our products by providing insights into our performance, baselines for customer expectations, and the ability to improve. MLCommons allows Lenovo to collaborate and engage with industry experts to create growth and in the end produce better products for our customers, which in today’s world is the number one focus here at Lenovo, customer satisfaction.

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MangoBoost

In the MLPerf Inference 6.0 submission round, MangoBoost is proud to announce groundbreaking results that push the frontier of LLM Inference Serving. Powered by our LLMBoost software, we achieved the first-ever submission utilizing multi-region GPU systems. Additionally, LLMBoost has continued to improve baseline performance, pushing our MI300X online results by adding 4% more throughput compared to our results from MLperf Inference v5.1.

Seamless Scalability in Complex Networks Using Dell's MI355X as the lead server, and MI325X and MI300X GPUs from MangoBoost's servers, LLMBoost orchestrated a heterogeneous mixture of multi-region GPU clusters. Despite serving on entirely separate networks across two regions (US and Korea) LLMBoost achieved near-perfect 94% performance scaling on llama2-70B across three simultaneous GPU generations. This configuration demonstrates unparalleled flexibility and scalability, serving as a powerful proof point for Dell hardware's and MangoBoost software's ability to maintain peak performance and effortless deployment, even within complex or pre-existing network architectures.

Maximizing TCO with Effortless Integration Unifying inference across disparate regions provides a transformative Total Cost of Ownership (TCO) advantage. This architecture enables geo-tied mapping, where the system intelligently routes users to the closest regional cluster to provide superior load balancing and minimal latency. Our results prove customers can easily integrate Dell's newest AMD-powered servers into existing pipelines using LLMBoost, scaling internationally without sacrificing efficiency or requiring massive network overhauls.

Turn-Key AI Infrastructure LLMBoost provides a turn-key solution bridging the gap from idea to production, allowing developers to serve open models at any scale in minutes. Beyond LLMBoost, MangoBoost offers advanced DPU-based hardware acceleration:

- **Mango BoostX™:** DPU for offloading networking, storage, and security workloads
- **Mango Storage Server:** High-density, disaggregated storage platform for high-bandwidth, low-latency NVMe access
- **Mango GPU Server:** Full-stack system purpose-built to maximize AI performance and tokens/$

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MiTAC

It is with great honor that MiTAC, a server platform designer, manufacturer, and a subsidiary of MiTAC Holdings Corporation (TWSE:3706), announces its outstanding results from the latest MLPerf® Inference v6.0 benchmark suite. These results validate our commitment to delivering cutting-edge AI infrastructure that pushes the boundaries of performance and efficiency. Our flagship G8825Z5 AI/HPC server series, including G8825Z5U2BC-350X-755 and G8825Z5U2BC-350X-575 powered by AMD MI350X GPUs, has demonstrated its prowess, with several performances in key Large Language Model (LLM) benchmarks.

The G8825Z5's remarkable achievements are a direct result of its purpose-built hardware and optimized design, which showcased in several categories. Standout performances include:

- LLaMA 2 70B Interactive: The G8825Z5 with AMD MI350X GPUs delivered impressive performance in the highly demanding interactive category, demonstrating its capability for real-time conversational AI.

- LLaMA 2 70B Server/Offline: The G8825Z5 showcased robust performance for high-throughput, latency-optimized server workloads.

These benchmark results affirm the G8825Z5's capability as a powerful solution for enterprises and research institutions seeking to deploy high-performance, real-time LLM applications at scale.

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Nebius

For MLPerf® Inference v6.0, Nebius submitted results for several NVIDIA Blackwell and NVIDIA Blackwell Ultra systems, including NVIDIA HGX B200, HGX B300, and GB300 NVL72. The benchmarking focused on evaluating these configurations for LLM and multimodal inference workloads across different cluster sizes. The HGX B200 and HGX B300 systems were tested on single-node configurations with 8 GPUs, while the GB300 NVL72 platform was evaluated across multiple setups, including single-node and multi-node configurations.

Nebius submitted results for three models: DeepSeek R1, Qwen3-VL 235B, and gpt-oss 120B. These widely used open-source models represent real-world AI workloads and are commonly used to evaluate the performance of modern inference systems. In addition, Nebius benchmarked the full rack-scale GB300 NVL72 system, including tests with the gpt-oss 120B model, providing insight into how advanced AI hardware platforms can support demanding inference workloads.

Nebius continuously tests, tunes, and optimizes its infrastructure to ensure the latest NVIDIA hardware platforms deliver high performance in a cloud environment. This work spans multiple layers of the AI infrastructure, from thorough server acceptance testing to cloud software optimization and multi-node cluster validation. These efforts help ensure stable performance and consistent system behavior across different hardware platforms, deployment sizes, and workload types.

The results of the MLPerf® Inference v6.0 submission demonstrate that the evaluated systems can reliably support demanding inference workloads, including disaggregated LLM serving. These outcomes highlight the readiness of Nebius's cloud infrastructure to run modern AI workloads while maintaining predictable and scalable performance.

The MLPerf® Inference benchmarks from MLCommons provide a standardized framework for evaluating AI inference performance across systems and platforms. Nebius's participation in MLPerf contributes to this effort and reflects the capability of its cloud infrastructure to support modern AI workloads with scalable and dependable performance.

Netweb Technologies India

Netweb Technologies announced its first submission to **MLCommons MLPerf™ Inference v6.0**, demonstrating the performance and readiness of its AI infrastructure platforms for enterprise and datacenter AI workloads. This milestone highlights Netweb's commitment to delivering high-performance, validated AI systems and participating in industry-standard benchmarking initiatives.

For the MLPerf Inference v6.0 Datacenter Closed division, Netweb benchmarked its **PDI200A2HG-810 AI server platform**, powered by **8× NVIDIA HGX B200-SXM-180GB GPUs** based on the latest Blackwell architecture. Each accelerator integrates **180 GB of HBM3e high-bandwidth memory** and is connected through **fifth-generation NVLink interconnects**, providing extremely high aggregate GPU-to-GPU bandwidth for large-scale inference workloads and memory-intensive models.

The submission includes results for modern AI workloads such as **automatic speech recognition using the Whisper model** and recommendation system inference using **DLRMv3**. These workloads represent key real-world enterprise AI scenarios including conversational AI, speech processing, and large-scale recommendation systems.

The platform is built around **dual Intel Xeon Platinum 8562 processors**, delivering a balanced CPU–GPU architecture with high host compute capacity and **2 TB of system memory** to support data preprocessing, scheduling, and inference orchestration. High-speed **PCIe Gen5 x16 connectivity** ensures efficient data transfer between CPUs and accelerators.

To support scalable AI deployments and cluster communication, the system integrates **400Gb/s InfiniBand interfaces**, enabling high-throughput networking suitable for distributed AI environments. The storage subsystem combines high-performance **NVMe SSDs** with network-attached storage to efficiently manage large datasets and AI model artifacts.

The MLPerf submission leveraged a modern AI software stack including **NVIDIA TensorRT, CUDA, cuDNN, TensorRT-LLM, NVIDIA Dynamo, and vLLM**, enabling optimized inference for large-scale models.

This first MLPerf participation highlights Netweb Technologies' ability to build **AI-optimized infrastructure platforms combining next-generation GPUs, high-bandwidth networking, and production-ready software frameworks**, reinforcing its commitment to transparent benchmarking and enterprise AI performance validation through MLCommons.

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NVIDIA

In MLPerf Inference v6.0, NVIDIA submitted many breakthrough results across a wide range of models using the NVIDIA GB300 NVL72 rack-scale system. In our best result, we used 4 such GB300 racks connected using Quantum-X800 InfiniBand, for a total of 288 Blackwell Ultra GPUs. As a result, NVIDIA achieved aggregate throughput of 2.5 million tokens per second on DeepSeek-R1.

Additionally, with the upgraded Blackwell Ultra GPU, GB300 NVL72 delivered up to 1.4x higher token throughput compared to optimized Blackwell GPU-based NVIDIA GB200 NVL72 results. Both GB200 NVL72 and GB300 NVL72 incorporate NVLink scale-up networking fabric, which connects all 72 GPUs together in a high-bandwidth all-to-all configuration designed to accelerate the latest models.

All low latency submissions on both GB200 NVL72 and GB300 NVL72 systems used NVIDIA Dynamo open-source software for disaggregated serving. NVIDIA submissions also used EAGLE-3 speculative decoding to accelerate the DeepSeek-R1 interactive scenario.

With just software optimizations, systems based on Blackwell Ultra increased token throughput by up to 2.3x in only six months. These continuous performance gains help drive down cost per million tokens with the same infrastructure, improving AI factory productivity.

Demonstrating the performance and versatility of the NVIDIA software stack, NVIDIA submitted excellent results on all newly-added models and their respective deployment scenarios. These included submissions on the more challenging DeepSeek-R1 interactive test with 5x higher interactivity compared to the server scenario, GPT-OSS-120B, Qwen3-VL (vision language model) and Wan 2.2 (text-to-video), as well as the new recommender benchmark, DLRMv3.

The NVIDIA partner ecosystem participated broadly, with 13 partners, including ASUS, Cisco, CoreWeave, Dell, GigaComputing, Google, HPE, Lenovo, Nebius, Quanta Computing, RedHat, Supermicro, and Lambda, demonstrating excellent performance.

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Oracle

OCI provides a comprehensive AI platform that includes AI Infrastructure, Generative AI services, Machine Learning services, and embedded AI across Oracle Fusion Applications. Customers can choose from a wide spectrum of GPU options tailored to different workload sizes. For entry-level use cases, OCI offers both bare metal and virtual machine instances powered by NVIDIA A10 GPUs. For mid-scale distributed training and inference workloads, customers can leverage NVIDIA A100 VM and bare metal instances, as well as NVIDIA L40S bare metal systems. At the high end, OCI supports the most demanding training and inference workloads with access to NVIDIA A100 80GB, H100, H200, GB200, and GB300 GPUs—scaling from single nodes to clusters with tens of thousands of GPUs. OCI offers AMD Instinct MI300X and MI355X accelerators as well, enabling advanced large-scale training and inference capabilities for open and proprietary models.

Recognizing that Generative AI workloads have fundamentally different performance characteristics and infrastructure requirements than traditional cloud applications, Oracle has engineered a purpose-built GenAI infrastructure stack. This includes a high-bandwidth, low-latency Cluster Networking architecture with Remote Direct Memory Access (RDMA) support—crucial for distributed training efficiency—as well as high-performance storage solutions powered by Managed Lustre. Together, these innovations ensure that OCI delivers the performance, scalability, and reliability needed to support the next generation of AI workloads.

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Quanta Cloud Technology

Quanta Cloud Technology (QCT), a global leader in data center solutions, continues to advance high-performance computing (HPC) and artificial intelligence (AI) with innovative system designs. In the latest **MLPerf™ Inference v6.0 benchmark**, QCT submitted results in the data center closed division, demonstrating diverse system configurations optimized for modern workloads.

- **QuantaGrid D75H-10U**: QCT Introduces the flagship and powerful 10U platform supporting up to eight NVIDIA HGX™ B300 GPUs with dual Intel® Xeon® 6 processors, 8x ConnectX®-8 OSFP ports serving 800G Ethernet or InfiniBand for 1:1 GPU-Direct RDMA. It eliminates data bottlenecks and delivers ultra-fast pipelines for large-scale AI inference and training.
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- **QuantaGrid D75E-4U**: A flexible 4U system supporting up to eight GPU accelerators with dual Intel® Xeon® 6 processors. This round we submitted the latest 8x RTX PRO 6000 on D75E-4U, enabling outstanding performance and flexibility for real-time inference and most demanding enterprise AI workloads.
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- **QuantaGrid D55Q-2U**: A high-density 2U server powered by dual Intel® Xeon® 6 processors for diverse workloads. With AMX instruction set support, it delivers optimized inference performance across multiple AI models such as llama3.1-8b, rgat, whisper, offering a cost-efficient alternative to GPU-based solutions.

By participating in MLPerf, the industry’s trusted AI benchmark, QCT underscores its commitment to validated performance, transparency, and customer confidence. These results empower enterprises, research institutions, and service providers to make data-driven infrastructure decisions with confidence.

Supplemental Results Discussion for MLPerf Inference v6.0

Red Hat

Red Hat, global provider of open source solutions, is proud to showcase MLPerf™ Inference results that demonstrate the breadth, flexibility, and performance of the Red Hat AI platform across diverse models, accelerators, and workloads.

This submission demonstrates the versatility of the Red Hat AI stack across real-world AI use cases and Red Hat's vision of empowering enterprises to deploy any model on diverse accelerators while seamlessly scaling to support more users and AI agents.

Our results span a diverse set of open models, including gpt-oss-120b, Qwen3-VL-235B-A22B, Llama2-70b, and Whisper evaluated on NVIDIA H200 and B200 GPUs and AMD Instinct MI350 GPUs, representing generative AI, multimodal vision-language, and speech models. Red Hat AI's inference stack is grounded on the leading open source inference solutions vLLM and llm-d. **vLLM** provides high-performance inference through efficient attention kernels, improved scheduling, asynchronous execution, mixed-precision and quantization optimizations to maximize accelerator utilization and throughput, while **llm-d**, a Kubernetes-native distributed LLM inference framework, enables intelligent load balancing and scalable deployments. Results were demonstrated using partner toolkits from NVIDIA and AMD—leveraging the vLLM runtime for some models—as well as Red Hat's own optimized inference harness, highlighting the flexibility of the Red Hat AI ecosystem.

Learn more about Red Hat AI and try out its integrated and optimized inference stack with Red Hat AI's Free 60-day trial.

Supplemental Results Discussion for MLPerf Inference v6.0

Stevens Institute of Technology

NVIDIA Blackwell Architecture and MoE kernel optimizations enable efficient inference

Yide Ran and Zhaozhuo Xu from Stevens Institute of Technology participated as individual members in this round of MLPerf® Inference and submitted results in collaboration with Lambda. The benchmarks were executed on Lambda's cloud infrastructure equipped with NVIDIA B200 GPUs featuring 180GB SXM memory, 112 CPU cores, 2.9 TB RAM, and 28 TB SSD.

Our benchmarking efforts include both the GPT-OSS-120B model and Llama 3 8B running on NVIDIA B200 GPUs for closed track. For the Open Division submission with GPT-OSS-120B, we explored kernel-level optimizations designed for mixture-of-experts (MoE) architectures, which help improve inference efficiency while maintaining model accuracy.

These results highlight the strong performance of Lambda's infrastructure for large-scale inference workloads. They also reflect a productive collaboration between Stevens Institute of Technology and Lambda, combining systems research expertise with high-performance cloud infrastructure to advance efficient LLM inference.

About Stevens Institute of Technology

Yide Ran is a PhD student and Zhaozhuo Xu is an assistant professor at Stevens Institute of Technology working on scalable and efficient machine learning systems. Their research focuses on high-performance AI infrastructure, efficient LLM inference and training, and algorithm–system co-design for large-scale machine learning.

Supplemental Results Discussion for MLPerf Inference v6.0

Supermicro

Supermicro is a global technology leader committed to delivering first-to-market innovation for AI, cloud computing, storage, and 5G/Edge. We are a Rack-Scale Total IT Solutions provider that designs and builds an environmentally-friendly and energy-saving portfolio of servers, storage systems, switches, software, along with global support services.

Supermicro offers optimized GPU systems to run AI inference on multiple platforms, including NVIDIA HGX B200/B300 GPU systems, AMD MI350X, MI355X GPU systems, and systems using the Intel Arc Pro B70 GPUs. Supermicro is the only vendor publishing datacenter closed benchmarks on the Intel Arc Pro B70 GPU.

Supermicro collaborated with others in this publication of MLPerf Inference v6.0. We had joint publication with Redhat, while Intel and AMD used Supermicro systems for their publication.

Supermicro offers a full range of systems with the latest GPUs from NVIDIA, AMD, and Intel, including NVIDIA GB200 and GB300 NVL72 systems.

Please go to https://www.supermicro.com for more information.